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Introduction to Computing for the Social Sciences

Session 10 – Machine Learning

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Winter Semester 2026-2027

Course Outline

- Session 1: Introduction
- Session 2: Information Coding
- Session 3: Data Structures
- Session 4: Software
- Session 5: Algorithms
- Session 6: Recursion
- Session 7: Sorting Algorithms
- Session 8: Time Complexity
- Session 9: Theoretical Computer Science
- Session 10: Machine Learning
  - Tutorial: Databases (swap)
- Session 11: Databases
- Session 12: Parallel Programming
  - Tutorial: Machine Learning (swap)
- Session 13: Social Science Applications
- Session 14: Exam Review

Outline

1. Brief History of Artificial Intelligence and Machine Learning
2. Supervised Machine Learning
3. Unsupervised Machine Learning
4. Generative AI and ML nowadays

Google predicted the flu, until it didn’t

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MOTIVATING EXAMPLE

In 2009, Google Flu Trends estimated influenza rates from search queries **two weeks ahead** of the CDC, and it worked. By the 2012–13 season it was predicting **more than double** the real rate. It had been fitted to millions of queries and had learned seasonal search patterns, winter, basketball, snow, rather than flu.

2009
ahead of the CDC, and celebrated

2×
the real rate, 2012–13 season

2015
quietly shut down

The model fitted the past almost perfectly. Why is that not a good sign, and how would you have caught it?

Ginsberg et al., *Nature* (2009); Lazer, Kennedy, King & Vespignani, “The Parable of Google Flu”, *Science* (2014)

93% accurate, and it never finds anyone

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MOTIVATING EXAMPLE

A university builds a model to flag students at risk of dropping out. It reports **93% accuracy** and is approved. Only 7% of students actually drop out, and the model has learned to answer **“no” every time**. Its accuracy is 93% and it identifies **nobody**.

93%
accuracy

0
students identified

0
recall, and therefore F_1

Which number should have been on the report instead, and what does the right answer depend on?

A million speeches nobody can read

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MOTIVATING EXAMPLE

You have every speech given in the Bundestag since 1949, on the order of a **million** of them. You want to know how the tone changed. Coding them by hand at five minutes each would take one person about **forty working years**. Coding a sample of **2 000** takes a week.

~10^6
speeches

~40 yr
to hand-code them all

2 000
you can actually code

How do you turn 2 000 human judgments into a measurement of a million documents, and how do you know whether to believe it?

Brief History of Artificial Intelligence

Alternative definitions of Intelligence:

a. Problem-solving ability of a system

b. “You know it when you see it” – The Turing Test

- **1950s:** Foundations in the US and Soviet Union
- **1970s:** The Perceptron Problem: first AI Winter
- **1980s:** Expert systems and the return of Neural Networks
- **1990s-2000s:** Second AI Winter
- **2010s:** Deep Learning thanks to compute and data
- **2020s:** The AI Bubble (TBD)



Artificial Intelligence (US)



Cybernetics (Soviet Union)

Machine Learning

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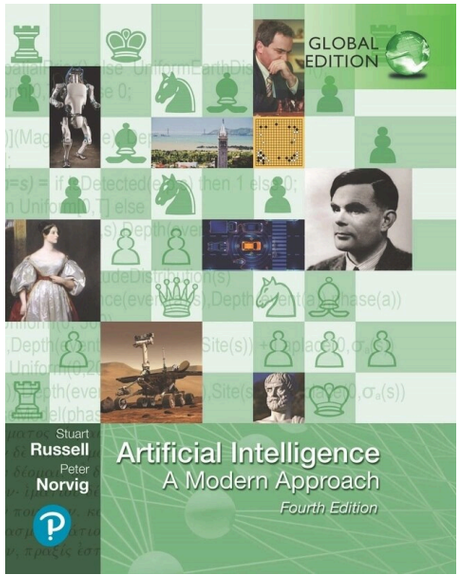
Two historical approaches to Artificial Intelligence:

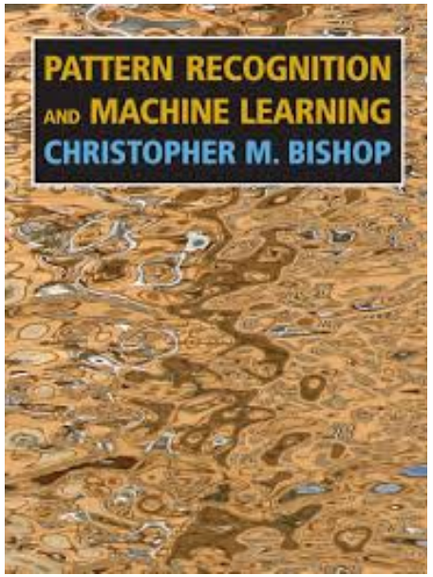
A) Expert systems that have hand-coded rules to solve one or more problems

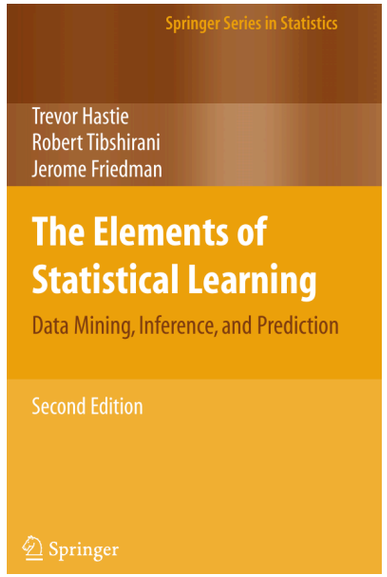
B) Machine Learning to automatically learn how to solve a problem from data

- Expert systems were dominant prior to the 80s and in the early 2000s, but now Machine Learning is the main approach to Artificial Intelligence
- Historically, Machine Learning has also been called “Pattern Recognition” and “Statistical Learning”
- Machine Learning trains a **Model** from data – the word “algorithm” is often used to refer to the resulting model
- In practice, AI systems might have expert rules but they mainly depend on learning from data

Readings in AI and Machine Learning







Recommended readings:

- Pattern Recognition and Machine Learning by Bishop
- The Elements of Statistical Learning by Hastie, Tibshirani and Friedman

Machine Learning Approaches

- **Supervised Learning: Learning with labeled examples**
  - Model learning from data where the correct answers are already known
  - e.g. predicting election outcomes based on historical voting data and demographics
  - Common tasks: Classification, Regression
- **Unsupervised Learning: Finding hidden patterns**
  - The algorithm explores data without labels to discover structure and relationships
  - e.g. grouping survey respondents into distinct segments based on their attitudes
  - Common tasks: Clustering, Dimensionality reduction
- **Reinforcement Learning: Learning through trial and error** (not covered today)
  - The algorithm learns by receiving rewards or penalties for its actions in an environment
  - e.g. adaptive language teaching or tests based on student performance
  - Common tasks: Sequential decision-making, Optimization

Key distinctions: supervised learning needs labeled data, unsupervised learning finds patterns on its own, and reinforcement learning needs access to an environment to learn from feedback

Supervised Machine Learning

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1. Brief History of Artificial Intelligence and Machine Learning
2. **Supervised Machine Learning**
3. Unsupervised Machine Learning
4. Generative AI and ML nowadays

Supervised Machine Learning

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The Goal

Learn a function $f : X \to Y$ that maps inputs to outputs, using a dataset of example input-output pairs.

Input Side

Features (X): input variables used to make predictions.
Also called: predictors, independent variables, covariates.
Example: age, income, education level

Feature Vector (x): a single observation with all its features.

```
x = [35, $50k, Bachelor's]
```

Training Data: collection of n examples:
$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$

Output Side

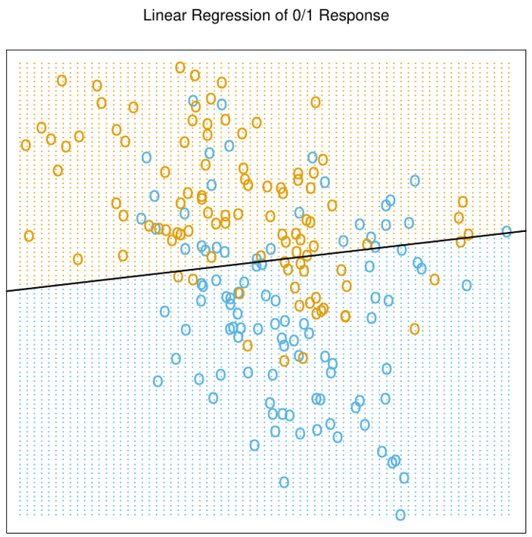
Target (Y): the outcome we want to predict.
Also called: label, response, dependent variable.
Example: voting choice, mortality

Prediction ($\hat{y}$): the model's output for a given input.
$\hat{y} = f(x)$

Model (f): the learned function that makes predictions

Classification and Regression

- **Classification:** Y is categorical, for example True/False or a label
  - Errors are either you get the right class or not
- **Regression:** Y is a continuous number, for example income or probability
  - Errors can be measured with a number



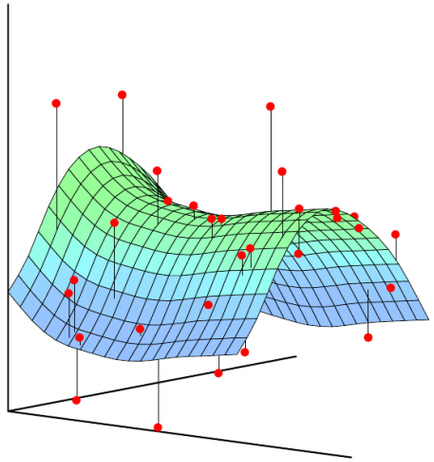


FIGURE 2.10. *Least squares fitting of a function of two inputs. The parameters of $f_\theta(x)$ are chosen so as to minimize the sum-of-squared vertical errors.*

Classification and regression examples from *Elements of Statistical Learning*

Model Features

- **Features** are the measurable properties or characteristics we use to describe our data. They are the input variables that the algorithm uses to learn patterns and make predictions.
- Think of features as the “questions we ask” about each data point. In social science: age, education, income. In text analysis: word counts, sentence length, sentiment scores.

Task: Classify movie reviews as **Positive** or **Negative**

Review 1 (Positive):

"This movie was amazing! Great acting and wonderful story."

Features (Word Frequencies):

amazing:	1	great:	1
wonderful:	1	terrible:	0
boring:	0	awful:	0

Review 2 (Negative):

"Terrible film. Boring plot and awful acting throughout."

Features (Word Frequencies):

amazing:	0	great:	0
wonderful:	0	terrible:	1
boring:	1	awful:	1

Feature Engineering: the process of creating, selecting, and transforming features to improve model performance. It’s about deciding *which* features to use and *how* to represent them. Examples: instead of just word counts, we might create features for word combinations (bigrams), the ratio of positive to negative words, or the presence of exclamation marks.

Features in Text Classification: TF-IDF

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The Problem with Simple Word Counts

Words like “the”, “and”, “is” appear frequently in every document but tell us nothing about sentiment. Meanwhile, distinctive words like “amazing” or “terrible” are very informative.

The TF-IDF Solution: TF-IDF (Term Frequency–Inverse Document Frequency) gives each word a score that balances how often it appears in a document with how rare it is across all documents.

- **Term Frequency (TF):** how often does the word appear in this document?

$$\text{TF} = \frac{\text{count of word}}{\text{total words in document}}$$

- **Inverse Document Frequency (IDF):** how rare is the word across all documents?

$$\text{IDF} = \log \frac{\text{total documents}}{\text{documents with word}}$$

- **TF-IDF score** of each word: $\text{TF-IDF} = \text{TF} \times \text{IDF}$

Words that are frequent in a document BUT rare overall get high scores.

Exercise: guess the highest and lowest TF-IDF scores in the example of the previous slide

Error and Loss

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Error

The difference between what the model predicted and the true value for a **single example**.

$$\text{Error} = \text{Prediction} - \text{True Value} \quad \text{or } \hat{y} - y$$

Loss

A function that measures the total error across **all examples** in the dataset. The goal of training is to minimize the loss.

$$\text{Loss} = \text{Average of errors (with different formulas)}$$

Review	True Label (y)	Prediction ($\hat{y}$)	Correct?	Error
“Amazing film!”	1 (Positive)	1 (Positive)	✓ Yes	1−1 = 0
“Terrible movie”	0 (Negative)	1 (Positive)	✗ No	1−0 = 1
“Loved it!”	1 (Positive)	1 (Positive)	✓ Yes	1−1 = 0
“Boring film”	0 (Negative)	0 (Negative)	✓ Yes	0−0 = 0
“Not good”	0 (Negative)	1 (Positive)	✗ No	1−0 = 1

Examples of Loss calculation:

Number of Errors: **2 out of 5**

Error Rate (0-1 Loss): **2/5 = 40%**

Accuracy: **60%**

In regression: Mean Squared Error, Mean Absolute Error. For probabilities: Cross-entropy

Class imbalance

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Class imbalance occurs when one class appears much more frequently than others in the dataset. This is extremely common in real-world problems.

Examples: fraud detection (99% legitimate, 1% fraud), rare disease diagnosis (95% healthy, 5% sick), hate speech detection (90% normal posts, 10% hate speech)

Accuracy is misleading under class imbalance

Example: Hate Speech Detection.

Dataset: 1,000 social media posts: 900 normal posts, 100 hate speech posts (10% imbalance)

Naive "Always Predict Normal" Model:

- Correct on normal posts: **900/900**
- Correct on hate speech: **0/100**
- Accuracy: **90%**

Looks great but detects ZERO hate speech!

The Issue: Accuracy treats all classes equally. A model can achieve high accuracy by just predicting the majority class every time, completely ignoring the minority class we actually care about!

The F1 Score

- **Precision**
  Of the posts we flagged as hate speech, how many actually were?

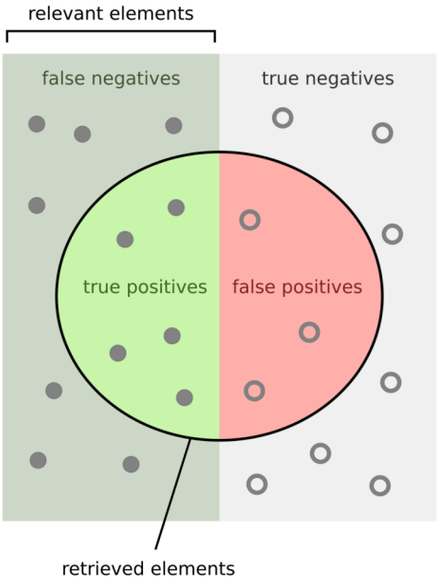
  `True Positives / (True Positives + False Positives)`
- **Recall**
  Of all actual hate speech, how many did we catch?

  `True Positives / (True Positives + False Negatives)`
- **F1 Score**
  Harmonic mean of precision and recall

  `2 × (P × R) / (P + R)`

In multi-class problems, the F1 Score can be calculated:

- for each class separately
- **Macro-average F1:** the mean of the F1 Score of each class
- **Micro-average F1:** a single score for all classifications together (larger classes weight more)



http://en.wikipedia.org/wiki/Precision_and_recall

Training/Test split - Bias and Variance

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The Ultimate Goal of (Machine) Learning: Generalization

We don’t just want a model that memorizes the training data. We want a model that performs well on **new, unseen data** from the real world. Generalization is what makes ML valuable.

To evaluate generalization, we split our data into two separate sets **before** training begins:

Training Set (70-80%) 📚

Used to train the model. The algorithm learns patterns from this data.

Like practice exams for students

Test Set (20-30%) 📝

Held back from training. Used only to evaluate final performance.

Like the final exam – never seen before

Critical Rule: the test set must remain completely separate and unseen during training!

Information leakage: when the train/test split is not well separated. **Typical sources of leakage:** using the test set for feature engineering, or ignoring sample correlations, for example power users or high-activity periods in social media data.

Cross-validation: partitioning into many train/test splits and reporting average performance

Model Example: k Nearest Neighbors

Classifying with k nearest neighbors in a 0/1 task:

$$\hat{Y}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

$N_k(x)$: the k nearest neighbors of x

Example for $k = 15$:

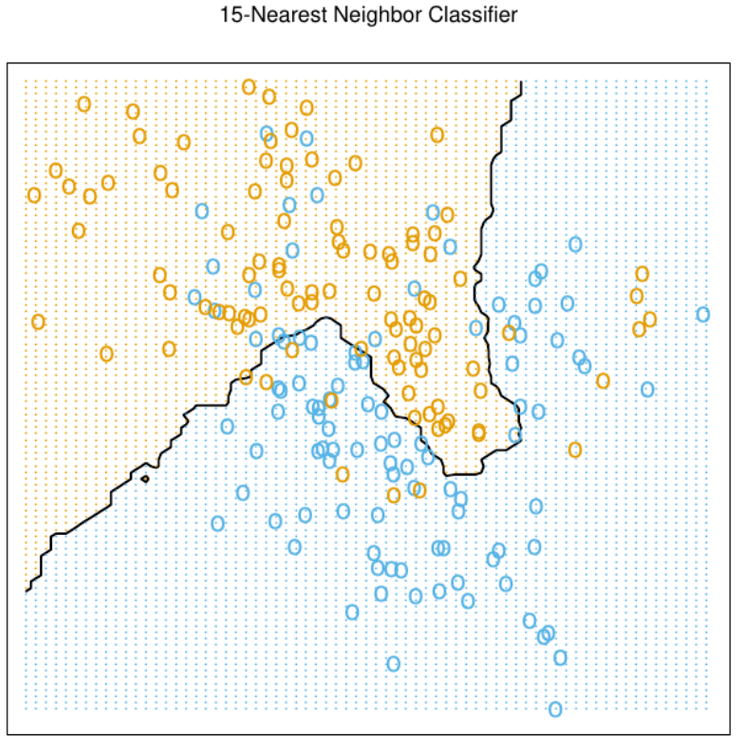


FIGURE 2.2. *The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable* (BLUE $= 0$, ORANGE $= 1$) *and then fit by* 15*-nearest-neighbor averaging as in (2.8). The predicted class is hence chosen by majority vote amongst the* 15*-nearest neighbors.*

Model Example: k Nearest Neighbors

Classifying with k nearest neighbors in a 0/1 task:

$$\hat{Y}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

$N_k(x)$: the k nearest neighbors of x

Exercise: extend the model for more classes

Example for $k = 1$:

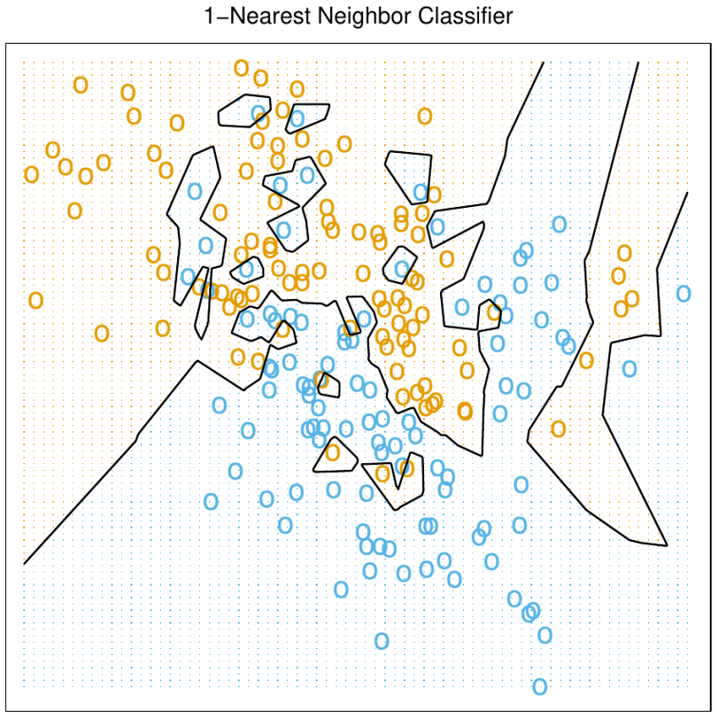
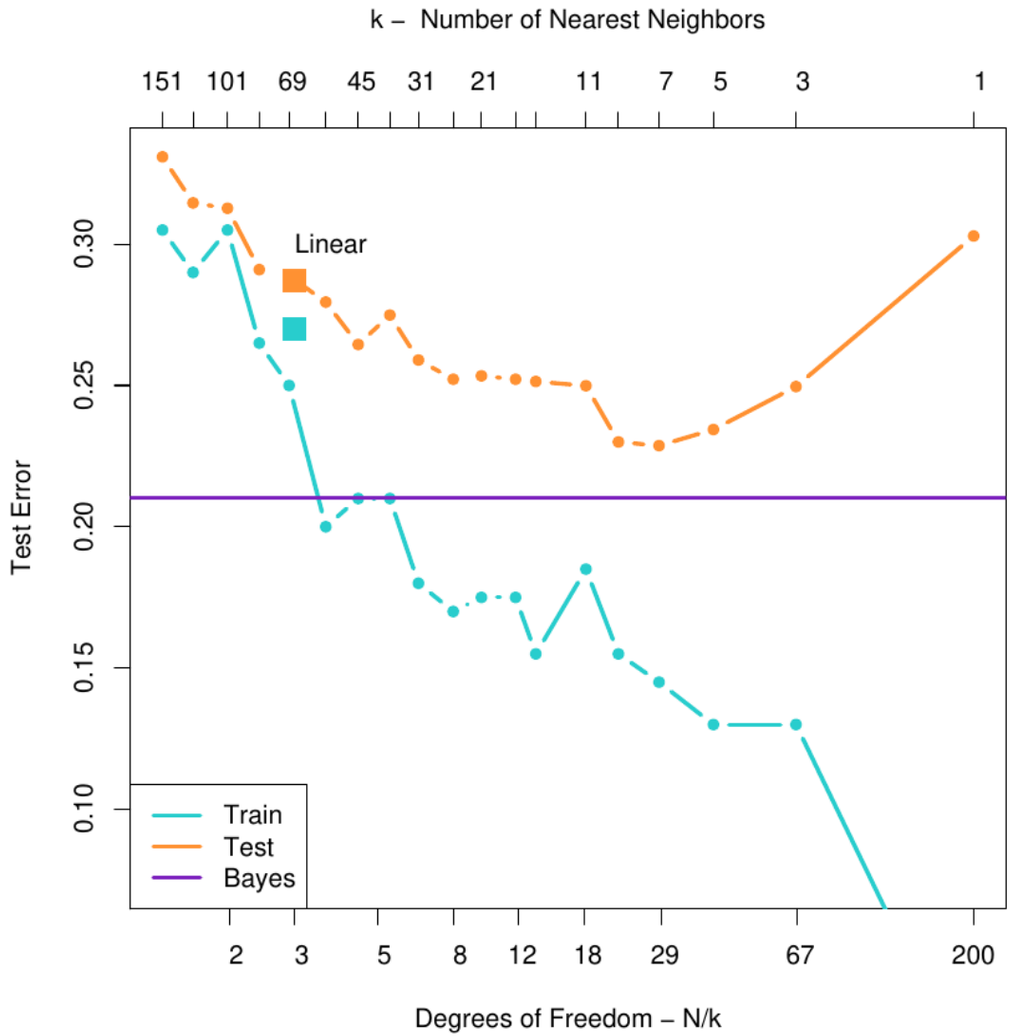


FIGURE 2.3. *The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable* (BLUE $= 0$, ORANGE $= 1$)*, and then predicted by* 1*-nearest-neighbor classification.*

Overfitting and Underfitting



- Misclassification curves for kNN
  - Training size: 200
  - Test size: 10000
- **Orange:** test error
- **Blue:** training error

- **Bias:** models with very high k (ca. 100) have high train and test errors – *underfitting!*
- **Variance:** models with very low k (ca. 1) have very low train error but high test error – *overfitting!*

The Bias-Variance Tradeoff

Consider **model complexity**

- e.g. number of parameters

Minimizing training error is a mistake that leads to high variance

Bias-Variance tradeoff: a middle range of complexity leads to the lowest possible test error

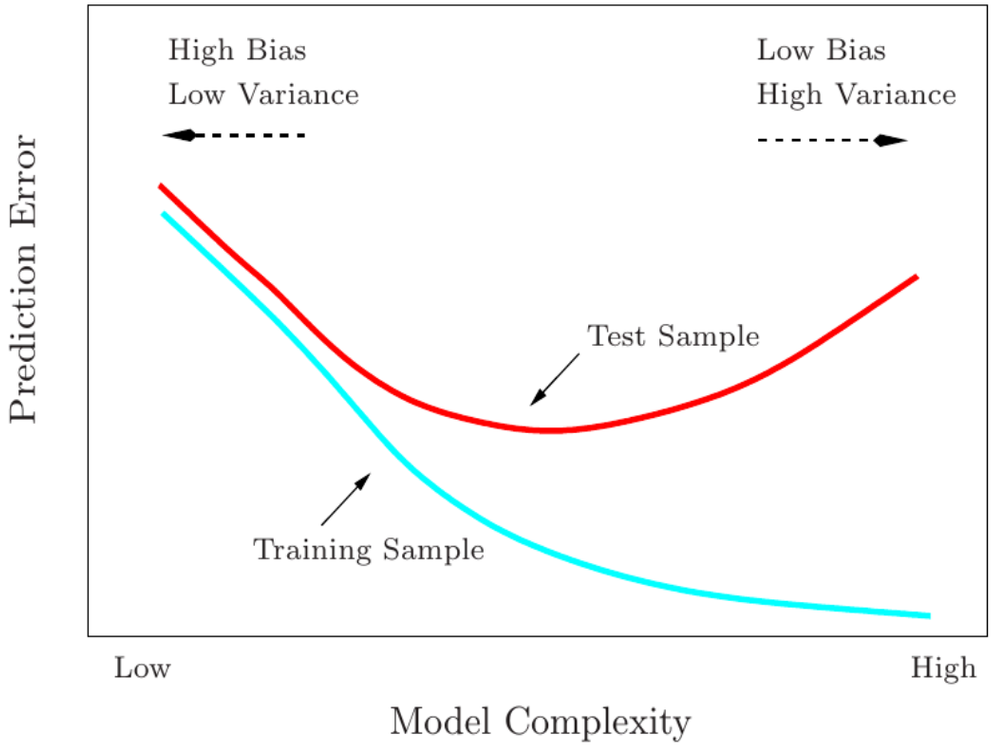


FIGURE 2.11. *Test and training error as a function of model complexity.*

Linear Model Example: Naive Bayes

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Naive independence assumption: features are independent of each other given the class, allowing us to decompose the joint probability into a product of individual feature probabilities.

Bayesian classification: for a feature vector $x = (x_1, x_2, \ldots, x_d)$, the predicted class is:

$$\hat{y} = \arg\max_{c \in C} P(Y = c \mid X = x) \quad \text{applying Bayes' theorem:}$$

$$\hat{y} = \arg\max_{c \in C} P(X = x \mid Y = c) \cdot P(Y = c)$$

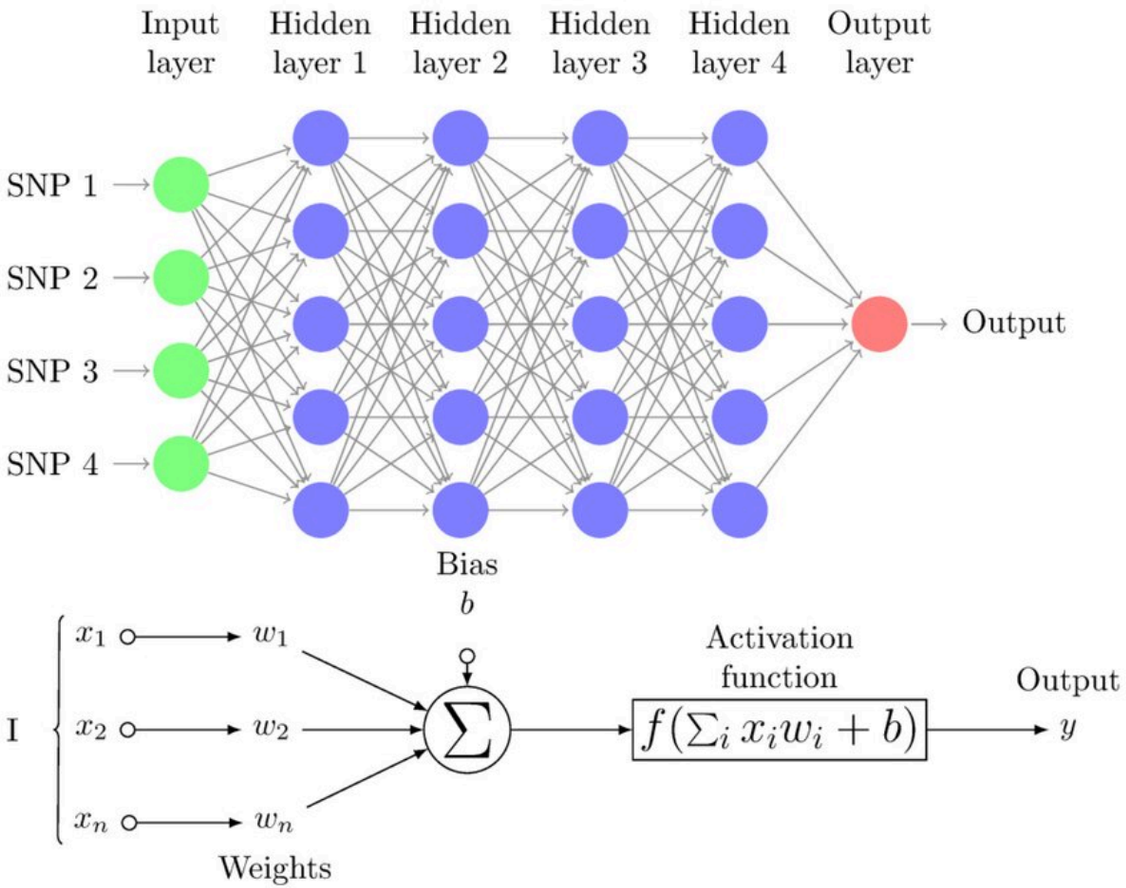
Under the naive independence assumption:

$$\hat{y} = \arg\max_{c \in C} P(Y = c) \cdot \prod_{j=1}^{d} P(X_j = x_j \mid Y = c)$$

- $\hat{y}$ = predicted class, Y = class variable, C = set of all possible classes
- $X = (X_1, X_2, \ldots, X_d)$ = feature vector with d features
- $P(Y = c)$ = prior probability of class c
- $P(X_j = x_j \mid Y = c)$ = likelihood of feature j having value x_j given class c

Naive Bayes is a **linear model** because it has a decision boundary that can be written as a hyperplane in the feature space X

Nonlinear Model Example: Multi-Layer Perceptron (MLP)



- Inspired by how neurons work: linear combination of inputs and nonlinear activation function
- **Features are the inputs of the first layer**
- The final output is a class or probability
- The complexity of the model grows with the number of hidden layers – more layers, more parameters
- **Fitting:** find connection weights that minimize loss
- **Nonlinear model:** it can fit more than just a hyperplane separating classes

Unsupervised Machine Learning

1. Brief History of Artificial Intelligence and Machine Learning
2. Supervised Machine Learning
3. **Unsupervised Machine Learning**
4. Generative AI and ML nowadays

Unsupervised Machine Learning

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Unsupervised Learning finds structures or patterns in data **without a target variable** that we try to predict.

In supervised learning we have pairs of feature vectors (X) and output or target values (Y), but in unsupervised learning we only have feature vectors (X).

Typical tasks for unsupervised learning:

- **Clustering:** partitioning the data into K groups where elements are similar within groups
- **Dimensionality reduction:** transforming X to a low-dimensional space maintaining similarity or variance (e.g. PCA)
- **Density estimation:** learning the underlying distribution of data values
- **Association discovery:** finding correlations or co-occurrence patterns

Clustering Example: Document Topics

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The Problem

Given a large collection of **unlabeled** documents (articles, tweets, research papers, survey responses), we want to automatically discover the main topics or themes and group similar documents together.

This is **unsupervised** because we don't know in advance what topics exist or which documents belong to which topic.



Clustering with K-Means

Algorithm 14.1 *K-means Clustering.*

1. For a given cluster assignment C, the total cluster variance (14.33) is minimized with respect to $\{m_1, \ldots, m_K\}$ yielding the means of the currently assigned clusters (14.32).

2. Given a current set of means $\{m_1, \ldots, m_K\}$, (14.33) is minimized by assigning each observation to the closest (current) cluster mean. That is,

$$C(i) = \underset{1 \le k \le K}{\operatorname{argmin}} ||x_i - m_k||^2. \tag{14.34}$$

3. Steps 1 and 2 are iterated until the assignments do not change.

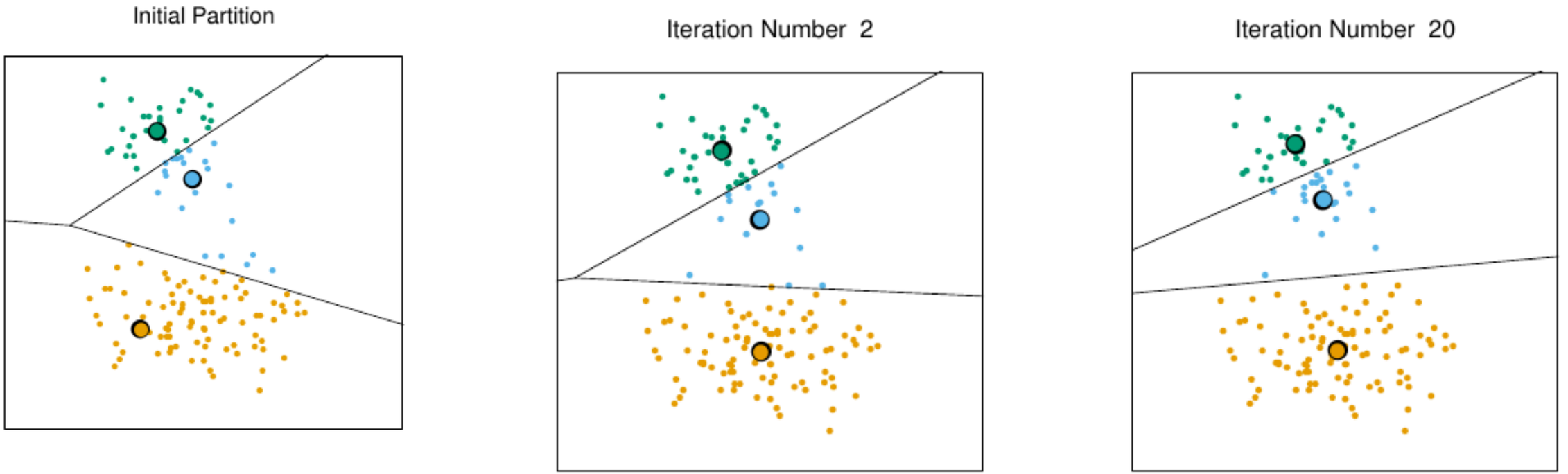
Mean of cluster S

$$\bar{x}_S = \underset{m}{\operatorname{argmin}} \sum_{i \in S} ||x_i - m||^2.$$

Total cluster variance

$$\min_{C, \{m_k\}_1^K} \sum_{k=1}^{K} N_k \sum_{C(i)=k} ||x_i - m_k||^2.$$

Clustering with K-Means



Examples from *Elements of Statistical Learning*

How to Choose K? The Elbow Method

We assume there is a “natural” number of clusters for K



K too small



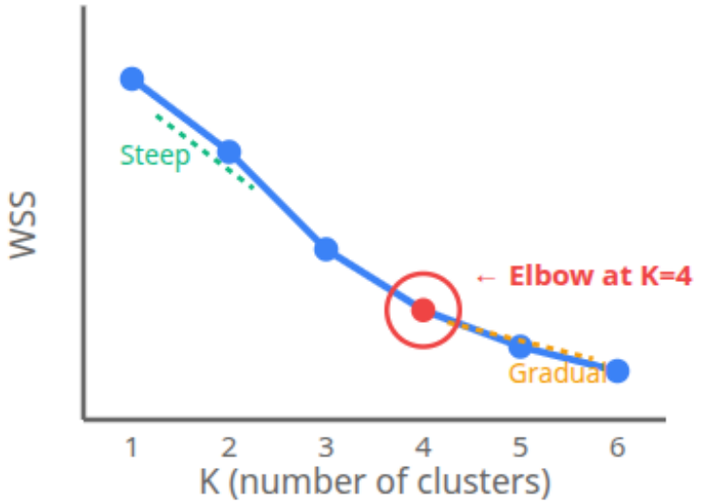
K just right



K too large

Within-Cluster Sum of Squares (WSS):

$$WSS = \sum_{k=1}^{K} \sum_{x_i \in c_k} ||x_i - \mu_k||^2$$



After the elbow, additional clusters barely reduce the WSS

Generative AI and ML nowadays

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4. **Generative AI and ML nowadays**

Predictive versus Generative Models

Predictive models aim to predict just one variable (Y), while **generative** models aim to produce new examples of data points (X).

Example for text:

Predictive (Discriminative)

Models the conditional probability of a label given the text:

$$P(y \mid x)$$

where y is the label/class and x is the input text

Example: $P(\text{sentiment} = \text{positive} \mid x)$ with x = “I loved this movie”

Generative

Models the joint probability or the probability of text:

$$P(x) \quad \text{or} \quad P(x_t \mid x_1, \ldots, x_{t-1})$$

Predicts the next token given previous tokens (**autoregressive**)

Example: $P(x_t = \text{movie} \mid \text{I loved this})$

Large Language Models

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Training Objective: Next-Word Prediction

Given a sequence of words, predict what word comes next. This is called **autoregressive language modeling**.

Mathematical Formulation:

$$\text{Maximize: } P(w_t \mid w_1, w_2, \ldots, w_{t-1})$$

Probability of word t given all previous words

Training Example:

Input: `"The cat sat on the"`

Target: `"mat"`

Model predicts:

mat: 0.65	floor: 0.20	table: 0.10	...: 0.05

Training Process (Simplified)

1. **Collect massive text data**
2. **Create training examples**
   For each position, use previous words as input, next word as target
3. **Train neural network**
   Update billions of parameters minimizing prediction error
4. **Repeat trillions of times**
   Process enormous amounts of text until the model learns language patterns

Note: some of today's slides were co-created with Claude. But this works thanks to the fact that I already know what Claude should say!

Geoffrey Hinton: The Godfather of AI



Geoffrey Hinton (born 1947) is a British-Canadian cognitive psychologist and computer scientist whose pioneering work on neural networks laid the foundation for modern AI, including generative models like ChatGPT, DALL-E, and Stable Diffusion.

- **2024 Nobel Prize – Physics** (for neural networks)
- **2018 Turing Award** – with Bengio and LeCun

Hinton describes AI as a kind of
intuition
rather than intelligence. Our intuition discovers patterns we cannot explain and that might not be useful for intervention or design. But it is a very useful tool for creativity and prediction.

Ethical issues with ML and Responsible AI

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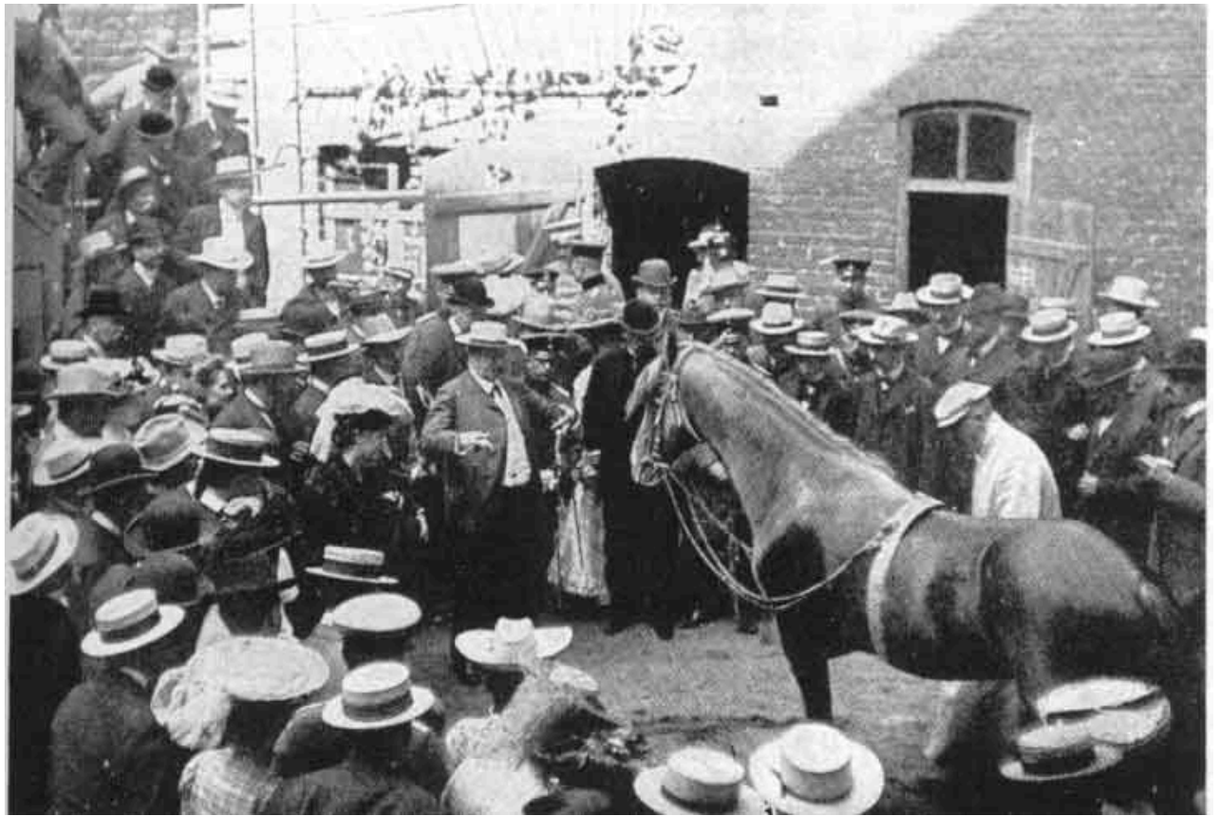
- **Can we use a model for a prediction?**
  - e.g. historical training data for unemployment benefits
- **What features are allowed for a prediction?**
  - e.g. against human rights to use ethnicity as a predictor for criminality
- **Who are we generalizing for?**
  - e.g. computer vision models not detecting non-white faces
- **Under which conditions was data gathered?**
  - e.g. text authors did not express agreement with it being used to train generative text models (labor loss)
- **Can we trust an AI model?**
  - people have a right to an explanation of a prediction (credit scores)
- **Who is responsible for model mistakes?**
- **Responsible AI:** not just preventing harm, but preventing AI from shifting power to the already powerful. Compels to AI development but safely.

Summary

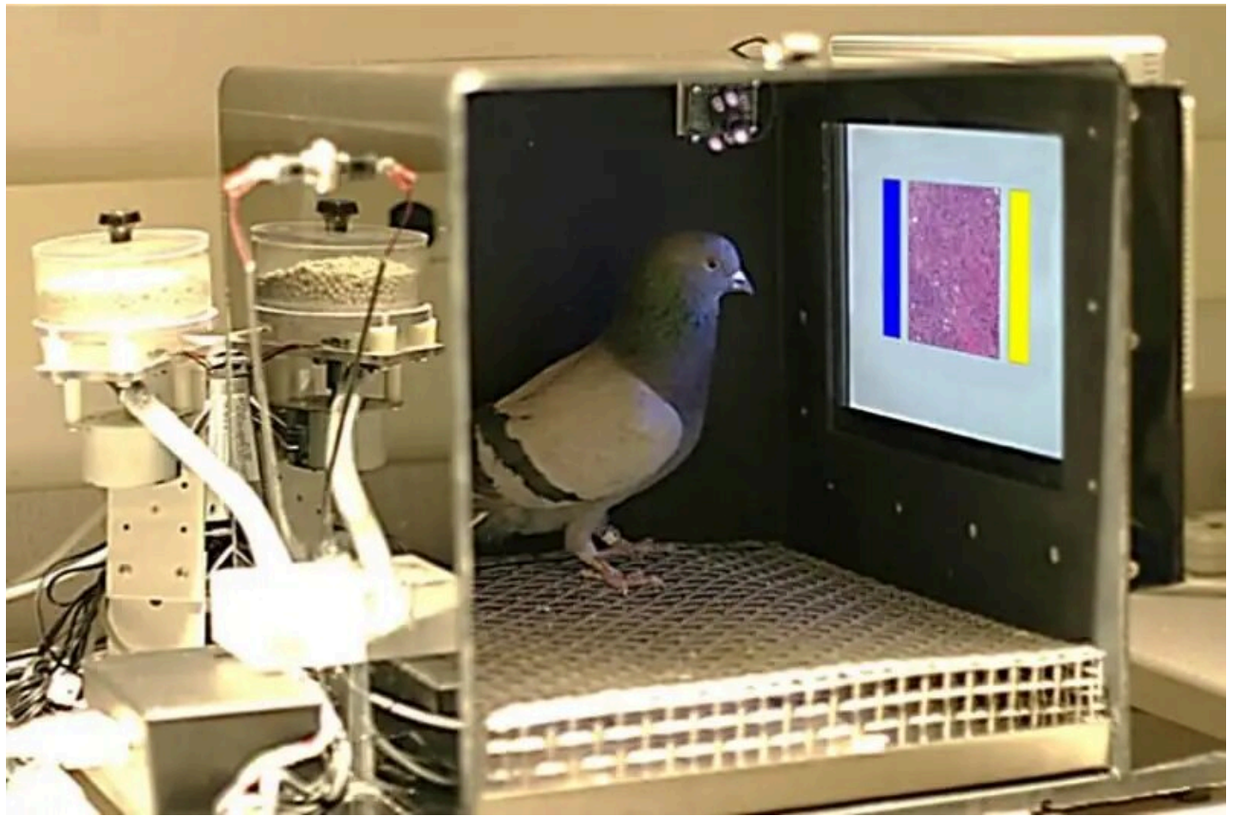
- **AI and Machine Learning**
  - Machine Learning as finding patterns from data
- **Supervised Machine Learning**
  - Predicting the value of a variable Y from features X
  - Bias-Variance Tradeoff, Train-Test split
  - Linear and nonlinear models: Naive Bayes and MLP
- **Unsupervised Machine Learning**
  - Finding patterns and structures in feature space X
  - Clustering as example – K-Means
- **AI nowadays:** Generative models, ML ethics and Responsible AI

Next session: Databases

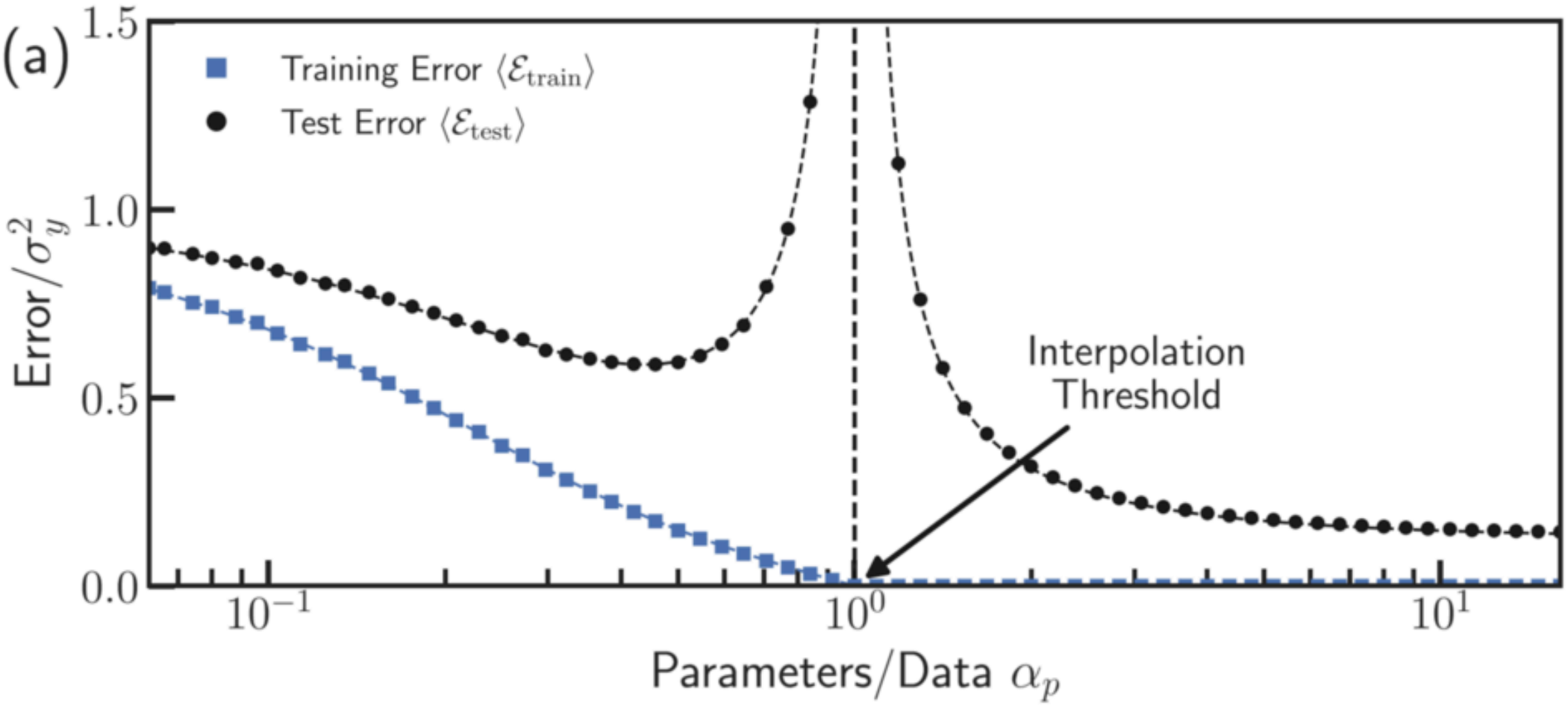
Appendix: Clever Hans



Appendix: Flocksourcing



Appendix: The Double Descent



Appendix: One Parameter is Always Enough

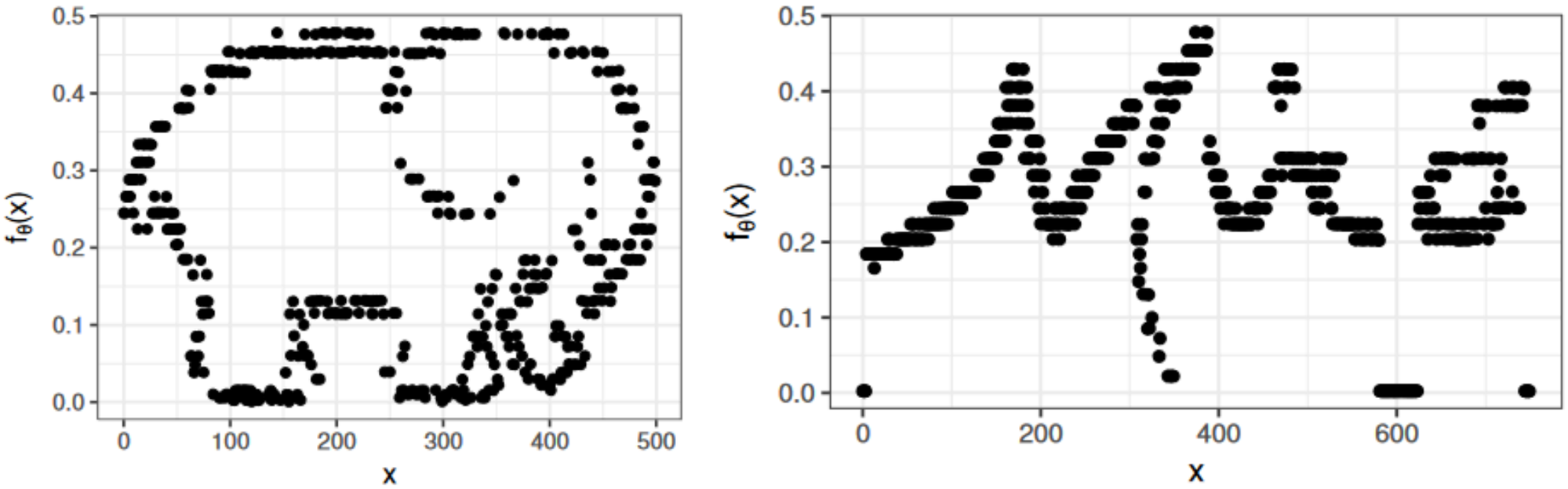
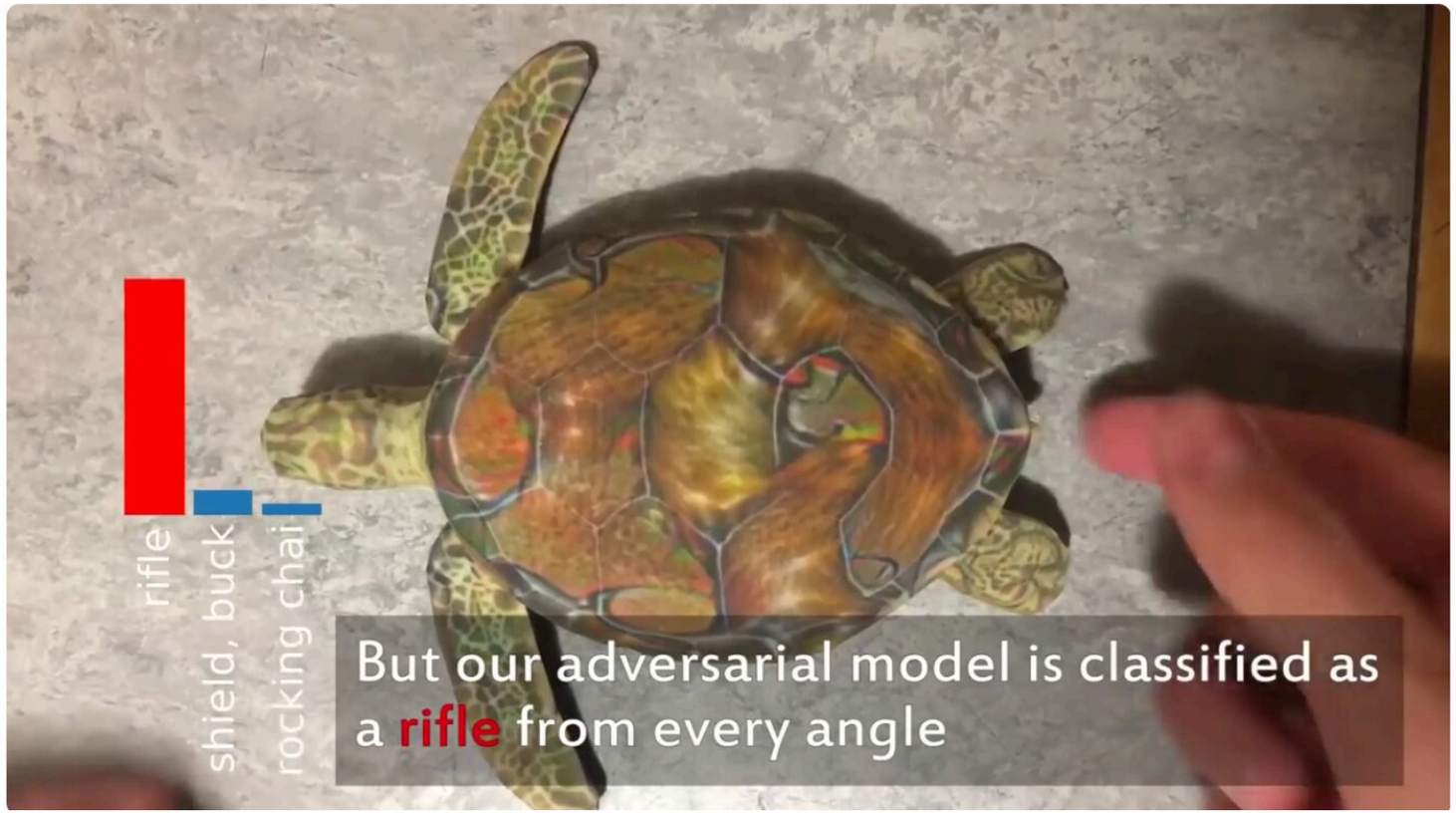


FIG. 1: A scatter plot of f_θ for $\theta = 0.24468472667347454582275406560\cdots$ plotted at integer x values, showing that a single parameter can fit an elephant (left). The same model run with parameter $\theta = 0.00242654180550004019353876200\cdots$ showing a fit of a scatter plot to Joan Miró's signature (right). Both use $r = 8$ and require hundreds to thousands of digits of precision in θ.

Appendix: Adversarial 3D-printed Turtle



Appendix: Microsoft Tay



Appendix: Fooling Gaydars



Appendix: Shadow Profiles

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